COMSATS University Islamabad,
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SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

for

RomaSub.AI:
Roman Urdu Captions Generator

By

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Bachelor of Science in Software Engineering (2022-2026)

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Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) ***include the ones given at scope time both in doc and presentation**	**Action Taken**

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Signature______________

1. Introduction

The Roman Urdu Subtitles (RomaSub.AI) system is designed to provide an automated solution for generating Roman Urdu subtitles from Urdu speech in audio or video content. This Software Requirements Specification (SRS) document defines the purpose, functional flow, system features, and constraints of the application. The system aims to address a significant accessibility gap for more than 50 million Urdu speakers who prefer Roman Urdu for digital communication but lack reliable tools for generating synchronized subtitles in this script.

With the increasing consumption of digital media, hearing-impaired individuals, educators, and content creators face challenges due to the absence of accurate and real-time Roman Urdu captioning tools. Existing platforms primarily support English or native Urdu scripts and require multi-step manual workflows. RomaSub.AI overcomes these limitations by integrating OpenAI Whisper for Urdu speech recognition and a Transformer-based transliteration engine to convert Urdu text into Roman Urdu with contextual accuracy.

Developed as a desktop application using Flutter for the frontend and FastAPI for backend operations, the system supports both real-time and offline subtitle generation, subtitle editing, user feedback integration, and multi-format export. The feasibility analysis indicates that the project can be built using open-source technologies with minimal cost, making it accessible for both academic and real-world use. This SRS outlines the architecture, scope, data requirements, processing methodology, and expected outcomes for the successful development of RomaSub.AI.

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the RomaSub.AI system - a desktop application designed to automatically generate Roman Urdu subtitles from Urdu speech in audio or video files. It serves as a foundation for the design, development, and validation of the system. The document ensures a clear understanding between stakeholders, developers, and users about what the system will achieve and how it will perform.

The main goal of RomaSub.AI is to provide users with an intelligent and efficient way to convert Urdu speech into accurately timed Roman Urdu subtitles. It aims to enhance accessibility for over 100 million Urdu speakers who use Roman Urdu for digital communication, particularly benefiting hearing-impaired individuals, content creators, and educators. The system emphasizes automation through advanced speech recognition and transliteration technologies while maintaining flexibility through offline processing, real-time subtitle display, and manual editing capabilities.

By clearly outlining system requirements, this document will help guide the development process and ensure that the final product aligns with user needs, institutional expectations, and technological constraints. The SRS will act as a reference for testing, maintenance, and future enhancements of the RomaSub.AI system.

1.2 Scope

RomaSub.AI is a desktop application designed to bridge the accessibility gap for more than 100 million Urdu speakers who use Roman Urdu for digital communication. It automatically converts Urdu speech from audio or video files into accurately timed Roman Urdu subtitles. The application includes secure user authentication, supports multiple media formats (such as MP4, AVI, MKV, and MP3), and uses OpenAI Whisper [2] for speech recognition along with a fine-tuned M2M100 model for AI-powered transliteration. Users can view subtitles in real time during video playback, make manual corrections through an interactive subtitle editor, and export subtitles in various formats including SRT, VTT, and TXT. Additionally, the system offers a feedback mechanism for continuous model improvement.

2. Overall description

The overall description section provides a comprehensive overview of the RomaSub.Ai, outlining its context, operational environment, and key technical constraints.

2.1 Product perspective

RomaSub.AI is a standalone desktop application designed to address a major gap in the Urdu language technology ecosystem by serving the large community of Roman Urdu users. Unlike existing subtitle generation platforms such as YouTube Auto-Captions, VEED.IO, and Maestra—which primarily support native Urdu script or widely used international languages—RomaSub.AI focuses specifically on generating subtitles in Roman Urdu. As an independent system, it functions entirely on the user’s machine without relying on external cloud services for its core processes. It accepts locally stored audio and video files as input and produces standard subtitle formats compatible with common media players. The application integrates advanced AI models, including OpenAI Whisper for speech recognition and a fine-tuned M2M100 model for transliteration, ensuring accurate and efficient subtitle generation.

2.2 Operating environment

RomaSub.AI will operate across multiple desktop platforms to ensure accessibility, stability, and a seamless user experience across devices. The desktop application will be developed using Flutter [3] to support Windows, macOS, and Linux operating systems, while the backend services will be built using FastAPI to ensure efficient processing and AI model integration.

OE-1: The system shall operate on Windows 10/11 (64-bit) desktop operating systems.

OE-2: The system shall operate on macOS 11.0 or later versions.

OE-3: The system shall require a minimum of 8GB RAM for basic operation and 16GB RAM for optimal performance with large video files.

OE-4: The system shall require at least 200mb to 350mb of available disk space for application installation, model storage, and temporary file processing.

OE-5: The system shall use PostgreSQL [1] 15.0 or higher for local database management.

The operating environment is designed for both development and scalability, ensuring optimal performance on local desktop systems while maintaining the flexibility for future enhancements.

2.3 Design and implementation constraints

The design and development of RomaSub.AI are influenced by specific technical, environmental, and academic constraints that define the tools, platforms, and frameworks used throughout the project. These constraints ensure consistency, maintainability, and compatibility across desktop platforms while keeping the project aligned with available university resources, computational capabilities, and academic timelines.

CO-1: The system shall be developed using Flutter 3.10+ for cross-platform desktop UI development to ensure consistent user experience across Windows, macOS.

CO-2: The system shall implement backend services using FastAPI 0.110+ framework to handle API requests and AI model integration efficiently.

CO-3: The system shall utilize OpenAI Whisper for Urdu speech recognition, as it provides the highest accuracy for Urdu language transcription among available open-source models.

CO-4: The system shall employ PyTorch 2.2 framework [4] for fine-tuning and inference operations due to its robust support for Transformer-based models.

CO-5: The system shall use Hugging Face [2] Transformers library for accessing and fine-tuning the M2M100 transliteration model.

CO-6: The system shall store all user data, feedback, and metadata in PostgreSQL 15.0 database to maintain data integrity and support complex queries.

CO-7: The system shall not implement cloud-based storage or processing to maintain user privacy.

CO-8: The system shall limit processing to single-speaker audio/video content in Version 1.0 due to complexity constraints of multi-speaker diarization.

CO-9: The system shall support only Urdu-to-Roman-Urdu conversion, excluding other language pairs to maintain focus and quality.

CO-10: The development shall utilize available hardware resources with optional Google Colab Pro access (budget: PKR 0-4,000) for model training tasks.

These constraints have been carefully defined to ensure that the system remains technically feasible, aligned with project goals, and adaptable for future upgrades beyond the academic scope.

3. Requirement identifying technique

Requirement Identifying Techniques are methods used to discover, gather, and document what users and stakeholders need from a software system. For RomaSub.AI, the primary technique used to identify and define system requirements is the Use Case Modelling approach. This method is highly effective for interactive end-user applications such as RomaSub.AI, which involves continuous user interactions across multiple modules including file upload, subtitle generation, real-time display, editing, and export functionality.

In addition to use cases, social media surveys and competitor analysis of existing tools like VEED.IO and Maestra were conducted to better understand user expectations and identify gaps in current subtitle generation solutions. The insights gathered were then converted into structured use cases and later expanded into detailed functional requirements in the subsequent section of this document.

3.1 Use case diagram

The following use case diagram illustrates the primary interactions between actors and the RomaSub.AI system:

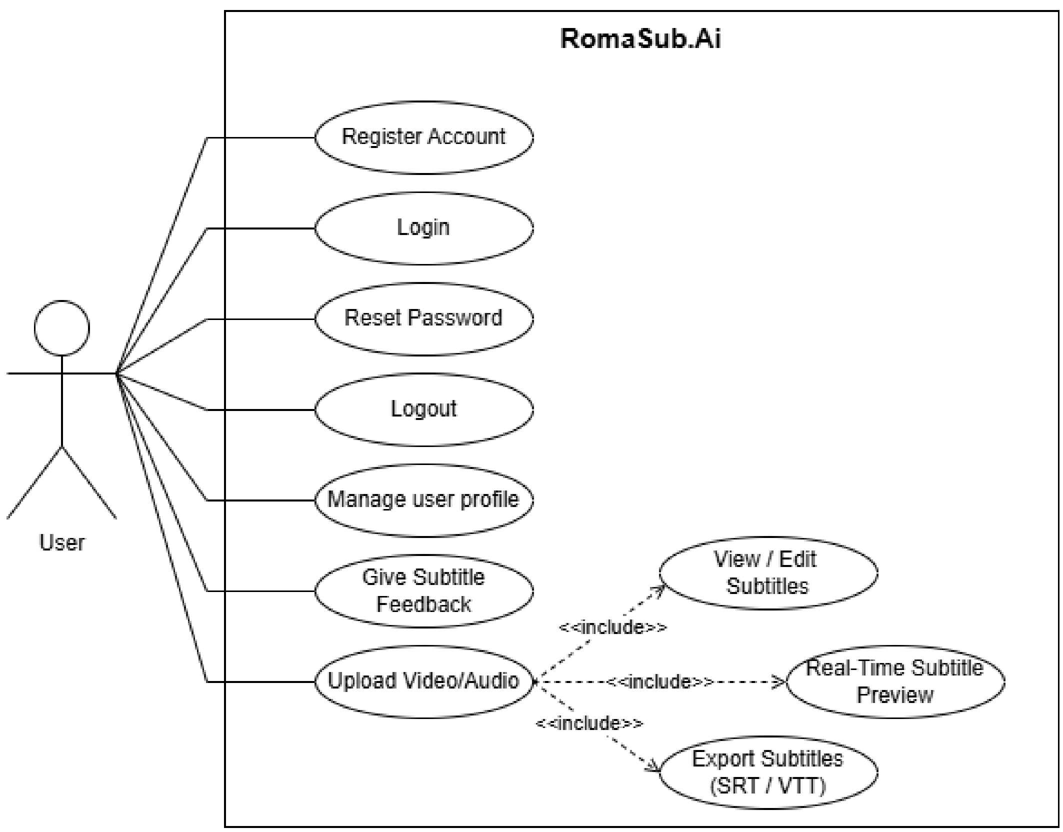

Figure 3.1 Use case diagram

3.2 Use case description

A fully dressed use case is a detailed, comprehensive specification that describes a complete interaction between an actor and system, including preconditions, main flow, alternative flows, post conditions, and exception handling.

UC-1 — Register Account

The following Table 1 gives the detailed description of the use case Register Account.

Table 1 UC-1 Register Account

Field	Description
Use Case ID:	**UC-1**
Use Case Name:	Register Account
Actors:	Primary Actor: User Secondary Actor: Authentication Service
Description:	This use case allows a new user to create an account by entering valid personal details. The system verifies the input and stores the user information securely.
Trigger:	The user selects the **"Register"** option from the main screen.
Preconditions:	• PRE-1: User must have an active internet connection. • PRE-2: Authentication API must be operational.
Postconditions:	• POST-1: User credentials are saved to the database. • POST-2: Confirmation message is displayed. • POST-3: User can now log in.
Normal Flow:	1. User clicks on **Register**. 2. System displays the registration form. 3. User enters full name, email, password. 4. System validates fields. 5. System checks if email already exists. 6. System stores user info. 7. System shows **Registration Successful** message.
Alternative Flows:	**1.1 Duplicate Email:** • System detects existing email.

	• User prompted to log in. **1.2 Weak Password:** • User enters weak password. • 2. System requests stronger password.
Exceptions:	• **E-1:** Network issue → retry option shown. • **E-2:** Server down → **“Service temporarily unavailable”** shown.
Business Rules:	• BR-1: Password ≥ 8 characters with letters + digits. • BR-2: Email must be unique.
Assumptions:	• User provides accurate personal details.

UC-2: Login

The following Table 2 gives the detailed description of the use case User Login.

Table 2 UC-2 Login

Field	Description
Use Case ID	UC-2
Use Case Name	Login
Actors	Primary Actor: Registered User Secondary Actor: Authentication Service
Description	This use case allows a registered user to log in to RomaSub.AI using their email and password. The system validates credentials and begins a secure session.
Trigger	The user selects the **Login** option from the main screen.
Preconditions	• PRE-1: User must already be registered. • PRE-2: Authentication service must be operational.
Postconditions	• POST-1: User session is created. • POST-2: User is redirected to the Dashboard.

Normal Flow	1. User opens the login screen. 2. User enters email and password. 3. System validates the input. 4. System verifies credentials in the database. 5. System authenticates and logs the user in. 6. User is redirected to the Dashboard.
Alternative Flows	**1. Incorrect Credentials:** • System detects mismatch. • System shows "Invalid email or password." **2. Remember Me:** • System stores a secure login token.
Exceptions	• **E-1:** Network failure during login → retry requested. • **E-2:** Service unavailable → error displayed.
Business Rules	• BR-1: Password must be securely hashed. • BR-2: Session must expire after defined inactivity.
Assumptions	A-1: User uses the correct registered email. A-2: System clock is accurate for session timing.

UC-3: Reset Password

The following Table 3 gives the detailed description of the use case Password Reset.

Table 3 UC-3 Reset Password

Field	Description
Use Case ID	UC-3
Use Case Name	Reset Password

Actors	Primary Actor: Registered User Secondary Actor: Authentication Service
Description	This use case allows a user to reset their password using an email verification code.
Trigger	User selects the **Forgot Password** option.
Preconditions	• PRE-1: User email must be registered. • PRE-2: Email verification service must be active.
Postconditions	• POST-1: Password is updated. • POST-2: User can log in using the new password.
Normal Flow	1. User selects **Forgot Password**. 2. System asks for the registered email. 3. User provides email. 4. System sends verification code. 5. User enters code. 6. User enters new password. 7. System validates and updates password.
Alternative Flows	**1. Wrong Verification Code:** • System prompts user to retry.
Exceptions	• **E-1:** Unregistered email → "No account found." • **E-2:** Verification service down.
Business Rules	• BR-1: Verification codes expire after a short period.
Assumptions	• A-1: User can access their email.

UC-4: Logout

The following Table 4 gives the detailed description of the use case Logout.

Table 4 UC-4 Logout

Field	Description
Use Case ID	UC-4
Use Case Name	Logout
Actors	Primary Actor: Authenticated User
Description	The user securely logs out, ending their session and returning to the login screen.
Trigger	User selects the **Logout** option.
Preconditions	• PRE-1: User must be logged in.
Postconditions	• POST-1: Session is terminated. • POST-2: User returned to Login page.
Normal Flow	1. User clicks **Logout**. 2. System asks to confirm if necessary. 3. System terminates the active session. 4. System navigates user to Login screen.
Alternative Flows	**1. Unsaved Work Warning:** • User chooses Save, Discard, or Cancel.
Exceptions	• **E-1:** Session token invalid → forced logout.
Business Rules	• BR-1: Logout must clear all tokens and cache.

Assumptions	• A-1: Logout completes within a few seconds.

UC-5: Manage User Profile

The following Table 5 gives the detailed description of the use case Manage User Profile.

Table 5 UC-5 User Profile

Field	Description
Use Case ID	UC-5
Use Case Name	Manage User Profile
Actors	Primary Actor: Authenticated User Secondary Actor: Database
Description	The user updates personal details such as name, password.
Trigger	User selects **Profile / Settings**.
Preconditions	• PRE-1: User must be authenticated.
Postconditions	• POST-1: Updated details saved.
Normal Flow	1. User opens the profile page. 2. User edits desired fields. 3. System validates input. 4. System updates database. 5. System shows success message.
Alternative Flows	**1. Change Password:** • User enters current password. • System validates → updates.

Exceptions	• **E-1:** Incorrect current password.
Business Rules	• BR-1: Email must be unique.
Assumptions	• A-1: User has stable internet (if cloud sync enabled).

UC-6: Provide Subtitle Feedback

The following Table 6 gives the detailed description of the use case Provide Subtitle Feedback.

Table 6 UC-6 Provide Subtitle Feedback

Field	Description
Use Case ID	UC-6
Use Case Name	Provide Subtitle Feedback
Actors	Primary Actor: Authenticated User Secondary Actors: Feedback System, Database
Description	User submits feedback to improve AI-generated subtitles.
Trigger	User selects **Submit Feedback**.
Preconditions	• PRE-1: Subtitles must be generated.
Postconditions	• POST-1: Feedback stored.
Normal Flow	1. User opens feedback form. 2. User enters comments/edits. 3. System validates input. 4. System saves feedback. 5. System displays acknowledgment.

Alternative Flows	**1. Training Contribution:** • User opts to include corrections in dataset.
Exceptions	• **E-1:** Empty form submission. • **E-2:** Database save failure.
Business Rules	• BR-1: Feedback must be linked to a project.
Assumptions	• A-1: User provides meaningful feedback.

UC-7: Upload Video/Audio File

The following Table 7 gives the detailed description of the use case Upload Video/Audio File.

Table 7 UC-7 Upload Video/Audio File

Field	Description
Use Case ID	UC-7
Use Case Name	Upload Video/Audio File
Actors	Primary Actor: Authenticated User Secondary Actors: File System, PostgreSQL Database
Description	The user uploads a video or audio file to begin the Roman Urdu subtitle generation process.
Trigger	User selects the **Upload File** or **New Project** option.
Preconditions	• PRE-1: User must be logged in. • PRE-2: User must have a video/audio file locally. • PRE-3: System must have enough disk space.
Postconditions	• POST-1: File is stored in the project directory. • POST-2: Audio is extracted if needed. • POST-3: Metadata is stored in the database. • POST-4: Project entry is created.

Normal Flow	1. User opens feedback form. 2. User enters comments/edits. 3. System validates input. 4. System saves feedback. 5. System displays acknowledgment.
Alternative Flows	**1. Cancel Upload:** • User cancels before confirmation. • System clears the selection and returns to Dashboard.
Exceptions	• **E-1:** Empty form submission. • **E-2:** Database save failure.
Business Rules	• BR-1: Feedback must be linked to a project.
Assumptions	• A-1: User provides meaningful feedback.

UC-8: View/Edit Subtitles

The following Table 8 gives the detailed description of the use case View/Edit Subtitles.

Table 8 UC-8 View and Edit Subtitles

Field	Description
Use Case ID	UC-8
Use Case Name	View and Edit Subtitles
Actors	Authenticated User
Description	User views and edits generated Roman Urdu subtitles by adjusting text and timing.
Trigger	User selects “Edit Subtitles” from preview screen or project menu.

<table>
<tr><td>Preconditions</td><td>Subtitles already generated.
User has previewed subtitles.</td></tr>
<tr><td>Postconditions</td><td>Subtitle edits saved to database.
Updated .srt file regenerated.</td></tr>
<tr><td>Normal Flow</td><td>1. User opens editor.
2. System shows video, subtitle list, and editor.
3. User selects a segment.
4. System loads segment details.
5. User edits text/timing.
6. User saves changes.
7. System updates database and marks segment modified.
8. User saves all and exits.</td></tr>
<tr><td>Alternative Flows</td><td>1. Add new subtitle segment.
• User clicks “Add Subtitle”.
• System creates a new subtitle segment.
• User edits timing/text and saves.

2. Delete subtitle segment.
• User selects a subtitle segment.
• User clicks “Delete”.

3. Undo / Redo change.
• User selects Undo or Redo.
• System reverts or reapplies the last edit.</td></tr>
<tr><td>Exceptions</td><td>• E1: Invalid time format.
• E2: Start time ≥ end time.
• E3: Timing overlap.
• E4: Text exceeds 500 characters.
• E5: Save to database failed.</td></tr>
</table>

Business Rules	• BR-1: Maximum characters per subtitle segment is 500. • BR-2: Subtitle duration must be between 0.5s and 7s. • BR-3: Minimum gap between segments must be 0.1s to prevent overlap. • BR-4: All edits must be logged with timestamp and user ID.
Assumptions	• A-1: User understands subtitle editing basics. • A-2: System has resources for undo/redo and saving.

UC-9: Real-Time Subtitle Preview

The following Table 9 gives the detailed description of the use case Real-Time Subtitle Preview.

Table 9 UC-9 Real-time subtitle preview

Field	Descriptions
Use Case ID	UC-9
Use Case Name	Real-Time Subtitle Preview
Actors	Primary Actor: Authenticated User Secondary Actors: Video Player Engine
Description	User views subtitles live during video playback.
Trigger	User clicks **Play Video** with subtitles loaded.
Preconditions	• Video and subtitles must be loaded.
Postconditions	• Subtitles are displayed in sync during playback.

Normal Flow	1. User loads video. 2. System loads subtitles. 3. User presses **Play**. 4. System synchronizes timestamped subtitles with video frames. 5. Subtitles appear on screen in real-time.
Alternative Flow	**1. User hides subtitles** • System removes on-screen captions.
Exceptions	• **E1:** Subtitle desync • **E2:** missing timestamps, • **E3:** unsupported video codec.
Business Rules	• BR-1: Subtitle display position must be centered horizontally at bottom 10% of video frame. • BR-2: Subtitle font size must auto-scale based on video resolution (12pt for 720p, 16pt for 1080p, 24pt for 4K). • BR-3: Subtitle background must have 70% opacity semi-transparent black overlay for readability. • BR-4: Fade-in and fade-out transitions must be exactly 200ms duration. • BR-5: Subtitle synchronization accuracy must be within ±50 milliseconds of speech timing. • BR-6: Minimum subtitle visibility duration is 0.5 seconds. • BR-7: Maximum subtitle visibility duration is 7 seconds.BR-46: In fullscreen mode, subtitle font size increases by 20% automatically.
Assumptions	• A-1: Hardware supports smooth video playback.

UC-10: Export Subtitles

The following Table 10 gives the detailed description of the use case Export Subtitles.

Table 10 UC-10 Export Subtitles

Field	**Descriptions**
Use Case ID	UC-10
Use Case Name	Export Subtitles
Actors	Primary Actor: Authenticated User Secondary Actors: File System
Description	User exports generated subtitles into a selected file format.
Trigger	User selects **Export** from the subtitle editor.
Preconditions	• Subtitles must be generated and/or edited.
Postconditions	• Subtitle file is saved locally.
Normal Flow	1. User clicks **Export**. 2. System shows format options (SRT, VTT, TXT). 3. User selects format and destination folder. 4. System validates path. 5. System exports the subtitle file. 6. System confirms export success.
Alternative Flow	**1. User cancels export** • System closes dialog without saving.

Exceptions	• **E1:** Invalid path • **E2:** insufficient permissions • **E3:** write failure.
Business Rules	• BR-1: Exported files must maintain subtitle timing accuracy within ±50ms of original timestamps. • BR-2: SRT and VTT formats must comply with respective international standards (SRT spec, W3C WebVTT). • BR-3: Export history must be maintained for 30 days in database for user reference. • BR-4: All exported subtitle files must use UTF-8 character encoding for Roman Urdu compatibility. • BR-5: File names must not contain special characters: / \ : * ? " < >
Assumptions	• User has write access to chosen directory.

4. Functional Requirements

This section lists the core functions that RomaSub.AI must perform, detailing the specific features and behaviours required to meet the needs of users and stakeholders.

4.1 User Registration

As shown in Table 1, FR-1 User Registration is mapped to its design component.

Table 1: FR-1 User Registration

Identifier	**FR-1**
Title	User Registration
Requirement	The user shall be able to register a new account by providing a valid full name, email, password, and password confirmation. The system shall verify email uniqueness before account creation.
Source	UC-1 (Register Account)

Rationale	To ensure secure and personalized access to user data and subtitle generation features.
Business Rule	BR-1: Passwords must contain at least 8 characters, one uppercase letter, and one number. BR-2: Email must be unique across all user accounts. BR-3: User data must be encrypted in the database.
Dependencies	FR-2 (Login)
Priority	High

4.2 User Login

As shown in Table 2, FR-2 User Login is mapped to its design component.

Table 2: FR-2 User Login

Identifier	**FR-2**
Title	User Login
Requirement	The system shall authenticate users using their registered credentials (email and password) and create a session token upon successful authentication.
Source	UC-2 (Login)
Rationale	To provide secure access to the system and enable user-specific features such as project management and recent history.
Business Rule	BR-1: Session tokens expire after 7 days of inactivity. BR-2: Maximum 3 failed login attempts within 15 minutes triggers 5-minute account lockout.
Dependencies	FR-1 (User Registration)
Priority	High

4.3 Reset Password

As shown in Table 3, FR-3 Reset Password is mapped to its design component.

Table 3: FR-3 Reset Password

Identifier	**FR-3**
Title	Reset Password
Requirement	The system shall allow users to reset their forgotten password by verifying their email address and providing a 6-digit verification code, then creating a new password that meets security requirements.
Source	UC-3 (Reset Password)
Rationale	To provide users with a secure mechanism to regain access to their accounts when they forget their passwords.
Business Rule	BR-1: Verification code must be exactly 6 digits. BR-2: Verification code expires after 15 minutes from generation. BR-3: New password cannot be same as previous password.
Dependencies	FR-1 (User Registration)
Priority	Medium

4.4 User Logout

As shown in Table 4, FR-4 User Logout is mapped to its design component.

Table 4: FR-4 User Logout

Identifier	**FR-4**
Title	User Logout
Requirement	The system shall terminate the user session, invalidate the session token, and redirect the user to the login screen when the user clicks the logout button. The system shall warn users of any unsaved changes before logout.
Source	UC-4 (Logout)
Rationale	To secure user accounts and prevent unauthorized access after user session ends.
Business Rule	BR-1: Logout must invalidate all active session tokens for the user. BR-2: System must warn user of unsaved changes before logout.
Dependencies	FR-2 (User Login)

Priority	High

4.5 Manage User Profile

As shown in Table 5, FR-5 Manage User Profile is mapped to its design component.

Table 5: FR-5 Manage User Profile

Identifier	**FR-5**
Title	Manage User Profile
Requirement	The system shall allow authenticated users to view and update their profile information including full name, email address, and password. The system shall validate email uniqueness and password strength before saving changes.
Source	UC-5 (Manage User Profile)
Rationale	To enable users to maintain current and accurate account information and enhance account security through password updates.
Business Rule	BR-1: Email changes take effect immediately without requiring verification in Version 1.0. BR-2: Password changes invalidate all existing session tokens except current session. BR-3: Full name must be between 3 and 100 characters.
Dependencies	FR-2 (User Login)
Priority	Medium

4.6 Provide Subtitle Feedback

As shown in Table 6, FR-6 Provide Subtitle Feedback is mapped to its design component.

Table 6: FR-6 Provide Subtitle Feedback

Identifier	**FR-6**
Title	Provide Subtitle Feedback

Requirement	The system shall allow users to submit feedback on ASR accuracy, transliteration quality, or general comments with a 1-5 star rating. The system shall optionally collect user edits as training data with explicit user consent, anonymize the data, and queue it for model fine-tuning.
Source	UC-6 (Provide Subtitle Feedback)
Rationale	To enable continuous improvement of AI models through user feedback and corrections, enhancing subtitle generation accuracy over time.
Business Rule	BR-1: User feedback must be stored for minimum 180 days for analysis. BR-2: Training data contributions must be anonymized before use in model fine-tuning. BR-3: Feedback with 2-star rating or below must be flagged as "Priority Review". BR-4: Users contributing training data receive double contribution points (2 points vs 1 point).
Dependencies	FR-10 (Generate Subtitles), FR-13 (Edit Subtitles)
Priority	Medium

4.7 Upload Video/Audio File

As shown in Table 7, FR-7 Upload Video/Audio File is mapped to its design component.

Table 7: FR-7 Upload Video/Audio File

Identifier	**FR-7**
Title	Upload Video/Audio File
Requirement	The system shall allow users to upload video (MP4, AVI, MKV) or audio (MP3) files via browse button or drag-and-drop interface. The system shall validate file format, size (maximum 2GB), duration (maximum 180 minutes), extract audio from video files, and create a project entry in the database.
Source	UC-7 (Upload Video)
Rationale	To enable users to input source media files for subtitle generation while ensuring system compatibility and resource constraints.
Business Rule	BR-1: Maximum file size supported is 2GB per upload. BR-2: Supported video formats are MP4, AVI, MKV with H.264/H.265 codecs.

	BR-3: Supported audio format is MP3 with bitrate between 64-320 kbps. BR-4: Maximum video duration supported is 180 minutes (3 hours). BR-5: Minimum disk space required is 3x the file size for processing.
Dependencies	FR-2 (User Login)
Priority	High

4.8 Real-Time Subtitle Preview

As shown in Table 8, FR-8 Real-Time Subtitle Preview is mapped to its design component.

Table 8: FR-8 Real-Time Subtitle Preview

Identifier	**FR-8**
Title	Real-Time Subtitle Preview
Requirement	The system shall display video/audio playback with synchronized Roman Urdu subtitles in real-time with accuracy within ±50 milliseconds. The system shall provide playback controls (play, pause, seek, volume, fullscreen) and subtitle toggle functionality.
Source	UC-8 (Real-Time Subtitle Preview)
Rationale	To enable users to preview generated subtitles synchronized with media content before editing or exporting.
Business Rule	BR-1: Subtitle display position must be centered horizontally at bottom 10% of video frame. BR-2: Subtitle font size must auto-scale based on video resolution. BR-3: Fade-in and fade-out transitions must be exactly 200ms duration. BR-4: Subtitle synchronization accuracy must be within ±50 milliseconds of speech timing.
Dependencies	FR-7 (Upload Video), FR-10 (Generate Subtitles)
Priority	High

4.9 Export Subtitles

As shown in Table 9, FR-9 Export Subtitles is mapped to its design component.

Table 9: FR- Export Subtitles in Multiple Formats

Identifier	**FR-9**
Title	Export Subtitles in Multiple Formats
Requirement	The system shall allow users to export subtitles in SubRip (.srt), WebVTT (.vtt), or Plain Text (.txt) formats. The system shall validate destination folder permissions, check disk space, handle file name conflicts, and save export records in the database.
Source	UC-9 (Export Subtitles)
Rationale	To enable users to export subtitles in standard formats compatible with various media players and video editing software.
Business Rule	BR-1: Exported files must maintain subtitle timing accuracy within ±50ms of original timestamps. BR-2: SRT and VTT formats must comply with respective international standards. BR-3: All exported subtitle files must use UTF-8 character encoding for Roman Urdu compatibility. BR-4: Users can export maximum 50 files per day to prevent resource abuse.
Dependencies	FR-10 (Generate Subtitles), FR-13 (Edit Subtitles - optional)
Priority	High

4.10 Generate Roman Urdu Subtitles

As shown in Table 10, FR-10 Generate Roman Urdu Subtitles is mapped to its design component.

Table 10: FR-10 Generate Roman Urdu Subtitles

Identifier	**FR-10**
Title	Generate Roman Urdu Subtitles
Requirement	The system shall process uploaded audio/video through OpenAI Whisper for Urdu speech recognition and M2M100 for Urdu-to-Roman Urdu transliteration. The system shall generate timestamped Roman Urdu subtitles, validate timing accuracy, and store results in database.
Source	Use case describing subtitle generation pipeline (ASR + Transliteration)

Rationale	Core functionality to automatically convert Urdu speech to Roman Urdu subtitles.
Business Rule	BR-1: ASR processing time shall not exceed 3x the duration of the audio file. BR-2: Minimum subtitle segment duration is 0.5 seconds. BR-3: Maximum subtitle segment duration is 7 seconds. BR-4: Minimum gap between subtitle segments is 0.1 seconds.
Dependencies	FR-7 (Upload Video), FR-11 (Whisper ASR Processing), FR-12
Priority	High

4.11 View and Edit Subtitles

As shown in Table 11, FR-11 View and Edit Subtitles is mapped to its design component.

Table 11: FR-11 View and Edit Subtitles

Identifier	**FR-11**
Title	View and Edit Subtitles
Requirement	The system shall provide a three-panel editor (video preview, subtitle list, text editor) allowing users to view, modify text, adjust timing, add new segments, delete segments, undo/redo changes, search subtitles, and auto-fix timing overlaps. The system shall validate all edits, auto-save every 30 seconds, and log changes for feedback system.
Source	UC-10 (View/Edit Subtitles)
Rationale	To enable manual correction and refinement of automatically generated subtitles, improving accuracy and quality of final output.
Business Rule	BR-1: Maximum 500 characters per subtitle segment. BR-2: Segment duration must be 0.5-7 seconds. BR-3: Minimum 0.1 second gap between segments. BR-4: All edits must be logged with timestamp. BR-5: Maximum 50 undo/redo operations. BR-6: Auto-save every 30 seconds.
Dependencies	FR-10 (Generate Subtitles), FR-8 (Real-Time Subtitle Preview)

Priority	High

5. Non-Functional Requirements

5.1 Usability

USE-1: The system shall enable a new user to successfully register, upload a video file, and generate subtitles within 10 minutes without external documentation or training.

USE-2: The system shall provide contextual error messages with suggested corrective actions for all user input validation failures.

USE-3: The subtitle editor shall allow users to make corrections with no more than 3 clicks or keystrokes per edit operation.

USE-4: The system shall maintain consistent UI design patterns (colors, fonts, button styles, layouts) across all screens to reduce learning curve.

USE-5: The system shall provide keyboard shortcuts for common operations (Save: Ctrl+S, Undo: Ctrl+Z, Play/Pause: Spacebar) to improve power user efficiency.

USE-6: The system shall display helpful tooltips when users hover over interface elements for more than 1 second.

5.2 Performance

PER-1: The system shall load and display the login screen within 3 seconds of application launch on systems meeting minimum hardware requirements.

PER-2: The system shall complete user authentication and navigate to dashboard within 2 seconds of successful credential submission.

PER-3: The system shall process ASR transcription at a rate no slower than 3x the duration of the audio file (e.g., 10-minute video processed in maximum 30 minutes) on systems with 8GB RAM and no GPU.

PER-4: The system shall complete transliteration of ASR output at a rate of minimum 100 words per second on systems meeting minimum hardware requirements.

PER-5: The system shall initiate video playback with subtitle display within 2 seconds of user clicking "Play" button.

PER-6: The system shall synchronize subtitle display with video frames with latency not exceeding 100 milliseconds.

PER-7: The system shall save subtitle edits to database within 1 second of user clicking "Save All" button for projects with up to 1000 subtitle segments.

PER-8: The system shall export subtitle files in any supported format within 5 seconds for projects with up to 3 hours of content.

PER-9: The system shall support simultaneous playback and subtitle editing with frame rate maintained above 24 FPS for 1080p video content.

5.3 Accuracy

ACC-1: The system shall achieve minimum 80% Word Error Rate (WER) accuracy for Urdu speech recognition on clear audio with minimal background noise.

ACC-2: The system shall achieve minimum 85% Character-Level Accuracy for Urdu-to-Roman Urdu transliteration on the provided test dataset.

ACC-3: The system shall maintain subtitle timestamp accuracy within ±50 milliseconds of actual speech timing.

ACC-4: The system shall correctly segment sentences with accuracy of minimum 90% based on natural speech pauses and punctuation.

ACC-5: The fine-tuned M2M100 model shall demonstrate measurable improvement (minimum 3% increase in Character-Level Accuracy) after training on 1000+ user-contributed correction samples.

5.4 Reliability

REL-1: The system shall maintain 99% uptime during normal operation hours, excluding scheduled maintenance.

REL-2: The system shall automatically recover from ASR or transliteration processing failures without requiring application restart.

REL-3: The system shall prevent data loss by auto-saving subtitle edits every 30 seconds during editing sessions.

REL-4: The system shall maintain data integrity by implementing database transaction rollback for failed operations.

REL-5: The system shall validate all subtitle files before export to ensure format compliance and prevent corrupted output.

REL-6: The system shall gracefully handle network interruptions during feedback submission by caching data locally and retrying when connection is restored.

5.5 Maintainability

MAIN-1: The system shall use modular architecture with clear separation of concerns between UI (Flutter), API (FastAPI), and AI model layers.

MAIN-2: The system shall include inline code documentation following language-specific documentation standards (Dart for Flutter, Python docstrings for FastAPI).

MAIN-3: The system shall maintain configuration files separate from source code to enable easy deployment environment changes.

6. References

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